

Estimating Extreme Credit Portfolio Risk with Importance Sampling

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1 Introduction

This project estimates the one-year credit loss risk of a loan or bond portfolio. Each issuer is characterized by a one-year default probability (PD), an exposure at default (EAD), and a loss given default (LGD). The main objective is to simulate possible portfolio losses and estimate tail-risk measures, including Value-at-Risk (VaR) and Expected Shortfall (ES).

A key feature of the model is that defaults are not assumed to be completely independent. Instead, a common systematic factor is introduced to represent the macroeconomic condition. When this factor is low, the economy is in a worse state, default events become more likely, and the portfolio loss may increase significantly. Therefore, this project uses a one-factor Gaussian credit model to describe the dependence among issuer defaults.

Plain Monte Carlo simulation is easy to implement, but it can be inefficient for estimating extreme credit losses. Because severe default losses are rare, most simulated scenarios correspond to normal market conditions, and only a small fraction of samples contribute to high-quantile risk measures such as 99% VaR and 99% ES. As a result, the estimates can be noisy unless the simulation size is very large.

To improve the efficiency of tail-risk estimation, this project applies Importance Sampling. The method deliberately generates more scenarios in adverse macroeconomic states, where large portfolio losses are more likely to occur. The bias introduced by this change of sampling distribution is then corrected using likelihood-ratio weights. Thus, Importance Sampling does not change the target risk measure; instead, it estimates the same VaR and ES more efficiently by increasing the number of informative tail-loss observations.

2 Data and Model Settings

In this study, the corporate credit portfolio is constructed by sampling issuers from two widely used U.S. corporate bond exchange-traded funds (ETFs): the iShares iBoxx Investment Grade Corporate Bond ETF (LQD) and the iShares iBoxx High Yield Corporate Bond ETF (HYG). These two ETFs are selected because they provide broad exposure to the U.S. corporate bond market across different credit quality levels.

LQD mainly consists of investment-grade corporate bonds, which generally have lower default risk and more stable credit performance. In contrast, HYG mainly consists of high-yield corporate bonds, which are non-investment-grade instruments with higher default risk but potentially higher yields. By sampling issuers from both ETFs, the constructed portfolio includes both relatively safe and relatively risky credit exposures. This makes the simulated portfolio diversified and representative of market-available corporate credit risk.

Exposure Breakdowns

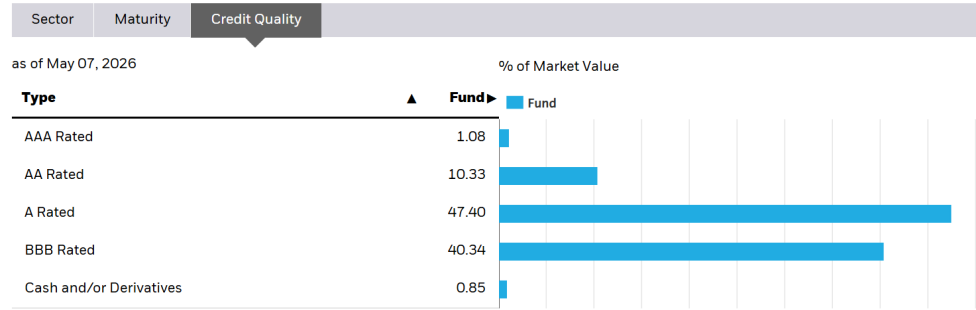


Figure 1: LQD overall rating distribution

Exposure Breakdowns

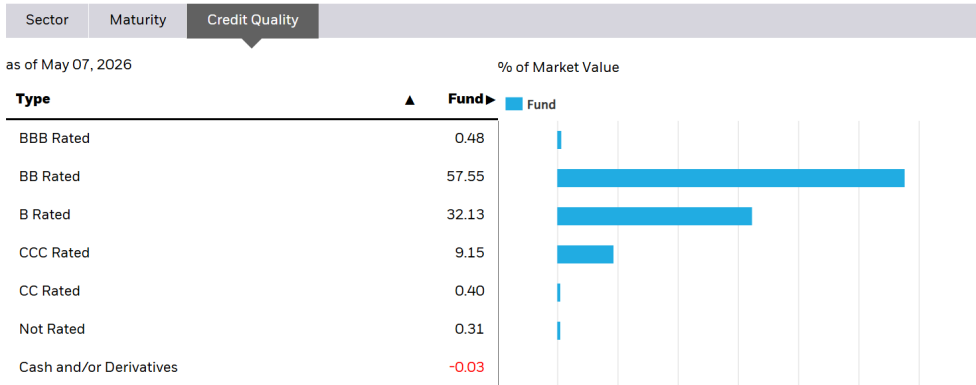


Figure 2: HYG overall rating distribution

After filtering out non-bond positions, the holdings are aggregated from the bond level to the issuer level using Bloomberg-based issuer identifiers. Issuers with missing default probability or exposure information are removed. We then form three 50-issuer portfolios with different LQD/HYG allocation splits: 30/70, 50/50, and 70/30. Within each ETF sleeve, issuers are selected using sector-stratified sampling based on market-value representation, and selected issuers are weighted proportionally to their issuer-level exposure. Finally, the sleeve weights are scaled to match the target LQD/HYG allocation, producing issuer-level portfolio weights used as inputs to the Monte Carlo credit-risk simulation.

Table 1: Portfolio Composition by ETF Source

Portfolio	LQD Companies	HYG Companies	LQD Weight	HYG Weight
30% LQD / 70% HYG	15	35	30%	70%
50% LQD / 50% HYG	25	25	50%	50%
70% LQD / 30% HYG	35	15	70%	30%

3 Model Setup

A one-factor Gaussian credit model is used in this project. For issuer i , the latent asset return is modeled as

$$X_i = \sqrt{\rho} Z + \sqrt{1 - \rho} \varepsilon_i, \quad (1)$$

where $Z \sim N(0, 1)$ is the systematic factor, $\varepsilon_i \sim N(0, 1)$ is the idiosyncratic shock, and ρ is the asset correlation. Issuer i defaults when

$$X_i < c_i, \quad c_i = \Phi^{-1}(p_i), \quad (2)$$

where p_i is the one-year default probability.

The portfolio loss in USD on each simulated path is defined as

$$L = \sum_{i=1}^n \text{EAD}_i \times \text{LGD} \times \mathbf{1}\{X_i < c_i\}.$$

Here, $\mathbf{1}\{X_i < c_i\}$ is the default indicator for issuer i . It equals 1 if issuer i defaults and 0 otherwise. Therefore, if issuer i defaults, its loss contribution is $\text{EAD}_i \times \text{LGD}$. In this project, all portfolios have 50 issuers, total exposure of \$100 million, loss-given-default $\text{LGD} = 0.4$, and asset correlation $\rho = 0.2$. Each simulation uses $N = 20,000$ paths, and each method is repeated $R = 50$ times.

The three portfolios differ in their risk composition. Their maximum one-year default probabilities are 0.1664, 0.1127, and 0.0307, respectively, so the first portfolio is the riskiest and the third portfolio is the safest.

4 Plain Monte Carlo and Importance Sampling

In plain Monte Carlo simulation, Z and ε_i are sampled from standard normal distributions, and portfolio losses are computed directly. The expected loss is estimated by the sample mean. The 99% VaR is estimated by the empirical 99% quantile of simulated losses, and the 99% ES is estimated by the average loss conditional on being above the estimated VaR:

$$\widehat{\text{ES}}_{0.99} = \frac{1}{\#\{j : L_j \geq \widehat{\text{VaR}}_{0.99}\}} \sum_{j: L_j \geq \widehat{\text{VaR}}_{0.99}} L_j. \quad (3)$$

In the importance sampling approach, only the distribution of the systematic factor is shifted:

$$Z \sim N(\mu_{\text{IS}}, 1), \quad \mu_{\text{IS}} < 0. \quad (4)$$

A negative shift creates more bad economic states and therefore more default-heavy scenarios. To keep the estimator unbiased for the original model, each simulated path receives the likelihood ratio weight

$$W(Z) = \frac{\phi(Z)}{\phi(Z - \mu_{\text{IS}})} = \exp\left(-\frac{Z^2}{2} + \frac{(Z - \mu_{\text{IS}})^2}{2}\right), \quad (5)$$

where ϕ is the standard normal density. Weighted quantiles and weighted tail averages are then used for VaR and ES.

The effective sample size (ESS) is also computed as

$$\text{ESS} = \frac{\left(\sum_{j=1}^N W_j\right)^2}{\sum_{j=1}^N W_j^2}, \quad (6)$$

which measures how much information remains after weighting. A more negative μ_{IS} usually creates more tail scenarios but also increases weight variability, so there is a tradeoff between tail exploration and ESS.

4.1 Choice of Importance Sampling Parameter

To select the shifting parameter μ_{IS} , a grid of candidate values from 0 to -1.4 was tested. For each candidate value, the standard error of ES₉₉ was estimated and compared with the effective sample size ratio. The final choice of μ_{IS} was based on a tradeoff between reducing ES99 variability and avoiding an excessively small effective sample size.

Table 2: Selected importance sampling parameters

Portfolio	Selected μ_{IS}	ES99 variance reduction
30/70 portfolio	-1.2	5.53×
50/50 portfolio	-1.3	8.04×
70/30 portfolio	-0.7	3.55×

5 Main Results

Table 3 summarizes the main numerical results. Only the expected loss, 99% VaR, and 99% ES are reported, since the focus of this project is extreme tail risk rather than the 95% risk level.

Table 3: Main results across 50 independent replications. Losses are reported in USD, assuming a total portfolio exposure of \$100 million.

Portfolio	Method	EL Mean	VaR99 Mean	ES99 Mean	ES99 SD	ES99 SE	Avg. ESS
30/70	Plain MC	428,153.19	3,578,162.10	4,519,998.95	107,257.63	15,168.52	–
30/70	IS	428,371.61	3,568,271.21	4,507,943.41	44,286.60	6,263.07	4835
50/50	Plain MC	277,378.51	2,748,532.05	3,568,087.58	97,781.05	13,828.33	–
50/50	IS	279,423.27	2,766,511.59	3,567,800.87	31,686.97	4,481.21	3740
70/30	Plain MC	147,970.77	2,110,395.02	2,651,535.66	58,477.97	8,270.03	–
70/30	IS	147,569.94	2,103,734.84	2,636,177.78	35,999.74	5,091.13	12236

The point estimates from plain Monte Carlo and importance sampling are close. This is important because it suggests that the likelihood ratio weighting is correctly adjusting for the shifted sampling distribution. The main difference is not the mean estimate itself, but the stability of the estimate across repeated simulations.

Table 4 focuses on the variability of the ES99 estimate. Importance sampling reduces the ES99 standard deviation from \$107,257.63 to \$44,286.60 for the 30/70 portfolio, from \$97,781.05 to \$31,686.97 for the 50/50 portfolio, and from \$58,477.97 to \$35,999.74 for the 70/30 portfolio. The corresponding variance reduction ratios are 5.53, 8.04, and 3.55.

6 Interpretation

The results show that importance sampling is most useful when the target statistic depends heavily on rare tail events. Expected loss is already estimated accurately by plain Monte Carlo because

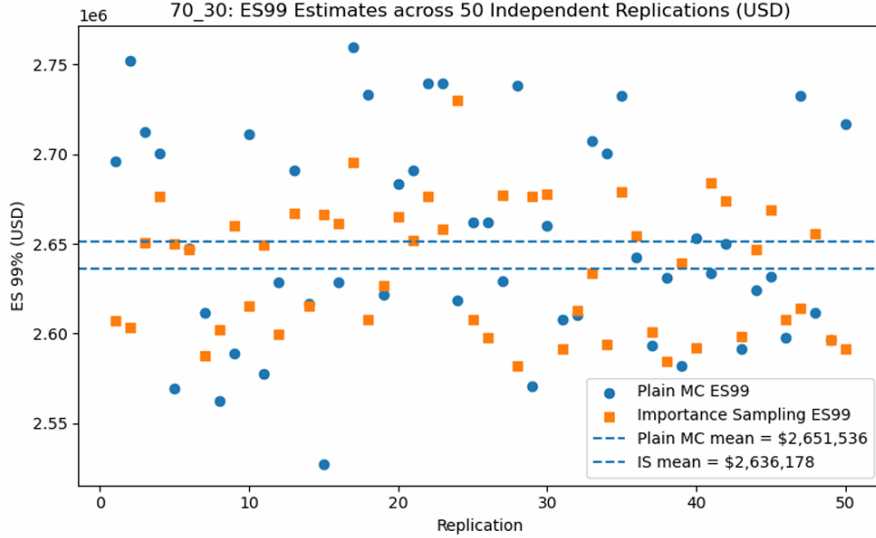
Table 4: Variance reduction in ES99 estimation.

Portfolio	μ_{IS}	Plain ES99 SD	IS ES99 SD	Variance Reduction	Avg. ESS
30/70	-1.2	107,257.63	44,286.60	5.53	4835
50/50	-1.3	97,781.05	31,686.97	8.04	3740
70/30	-0.7	58,477.97	35,999.74	3.55	12236

it averages over the entire loss distribution. In contrast, ES99 depends only on the worst 1% of outcomes, so plain Monte Carlo uses relatively few informative observations. By shifting the systematic factor left, importance sampling produces more high-loss scenarios and therefore makes the ES99 estimate more stable.

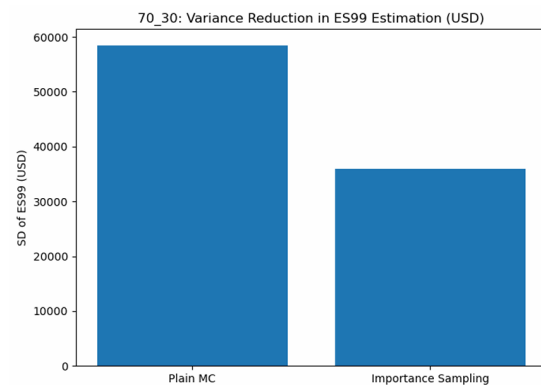
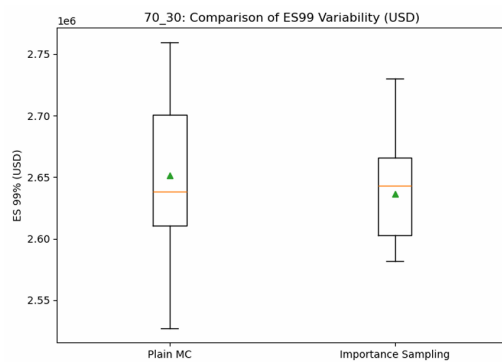
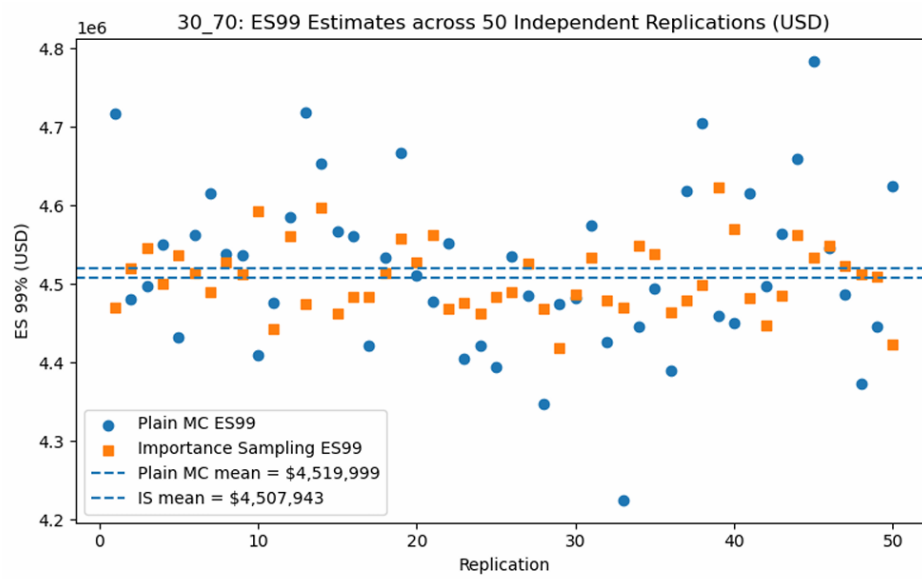
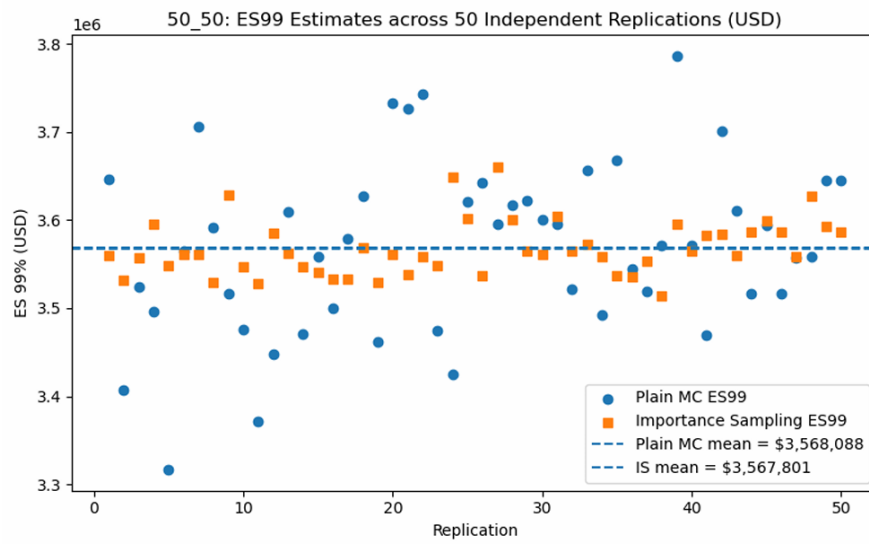
The second portfolio obtains the largest variance reduction ratio, 8.04. This suggests that the selected shift $\mu_{IS} = -1.3$ creates an effective balance between generating enough tail losses and avoiding excessive likelihood-ratio variability. The first portfolio also benefits strongly, with a variance reduction ratio of 5.53. The third portfolio still improves, but the gain is smaller at 3.55. One possible explanation is that the third portfolio has lower default probabilities overall, so even after shifting the systematic factor, extreme losses remain relatively sparse.

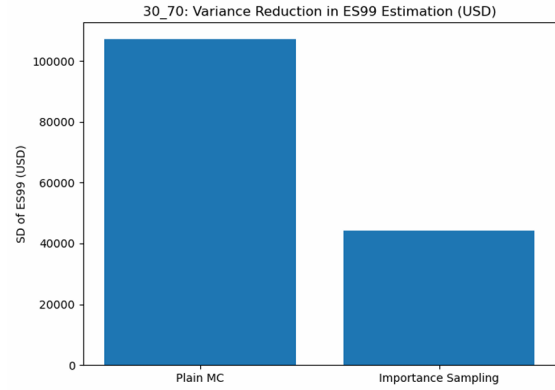
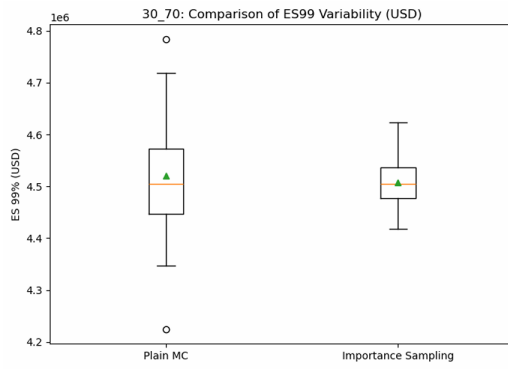
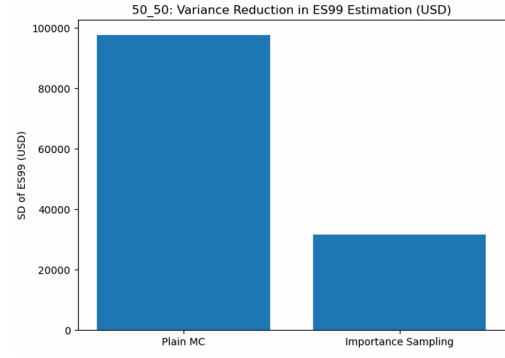
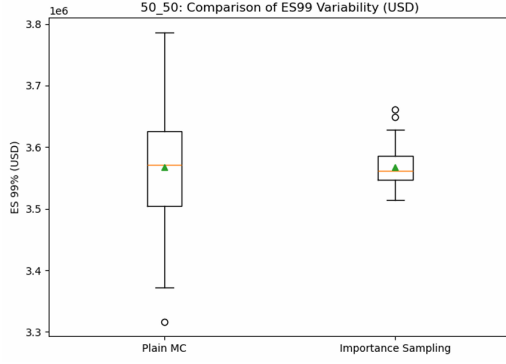
The ESS values also highlight the cost of importance sampling. The shifted simulations no longer contribute equally; some paths receive larger weights than others. For example, the average ESS is 4835, 3740, and 12236 out of 20,000 for the three portfolios. These values are much lower than the nominal simulation size, but the tail-risk estimates are still more stable because the simulation budget is concentrated in the most relevant part of the loss distribution.



7 Conclusion

This project demonstrates that importance sampling can substantially improve the efficiency of Monte Carlo estimation for extreme credit portfolio losses. For the three portfolios considered, the method gives similar EL, VaR99, and ES99 point estimates compared with plain Monte Carlo,





while reducing the variance of the ES99 estimator by factors of 5.53, 8.04, and 3.55. The main takeaway is that changing the sampling distribution of the systematic risk factor is an effective way to obtain more reliable tail-risk estimates without increasing the total number of simulations.

At the same time, the choice of μ_{IS} matters. If the shift is too small, the method does not create enough tail scenarios; if it is too large, the likelihood ratio weights become unstable and the effective sample size falls. A useful extension would be to choose μ_{IS} automatically by minimizing the estimated ES99 standard error subject to a minimum ESS constraint.

References

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